OVERVIEW

The finished pedal: 400 × 288 × 52 mm, two printed parts (base + top plate), the board and the Raspberry Pi inside.

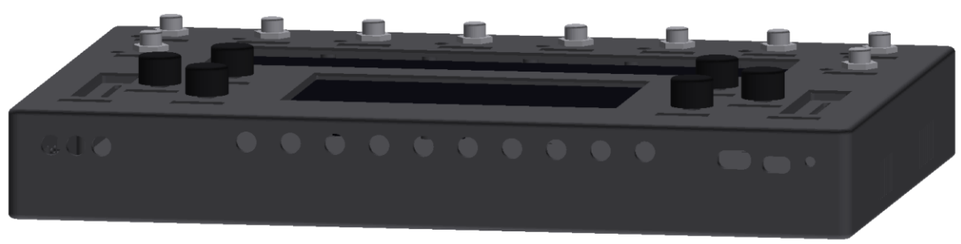
assembled



right side: Ethernet / USB / USB / HDMI 2 panel ports



back panel: power, USB-C, USB-A, jacks, MIDI, phones



WHAT IS IN THE BOX

Base (printed, 4 tiles): floor, walls, back panel, right I/O panel, Pi bosses with M2.5 inserts, fan pads, board bosses.

Top plate (printed, 4 tiles): every control comes through it - footswitch nuts on top, knobs on 30 mm shafts,

fader caps in recessed trays, bezelled windows for the 7" and 11.9" screens and the 18 label screens.

Main board rev B.3 (PCBWay, assembled): all controls, jacks, label screens; the 7" and 11.9" screens screw onto it.

Raspberry Pi 4 on the base floor, component side up; the board plugs onto it through a 2×20 stacking header.

40 mm fan on the floor over the exhaust grille; side vents are the intake.

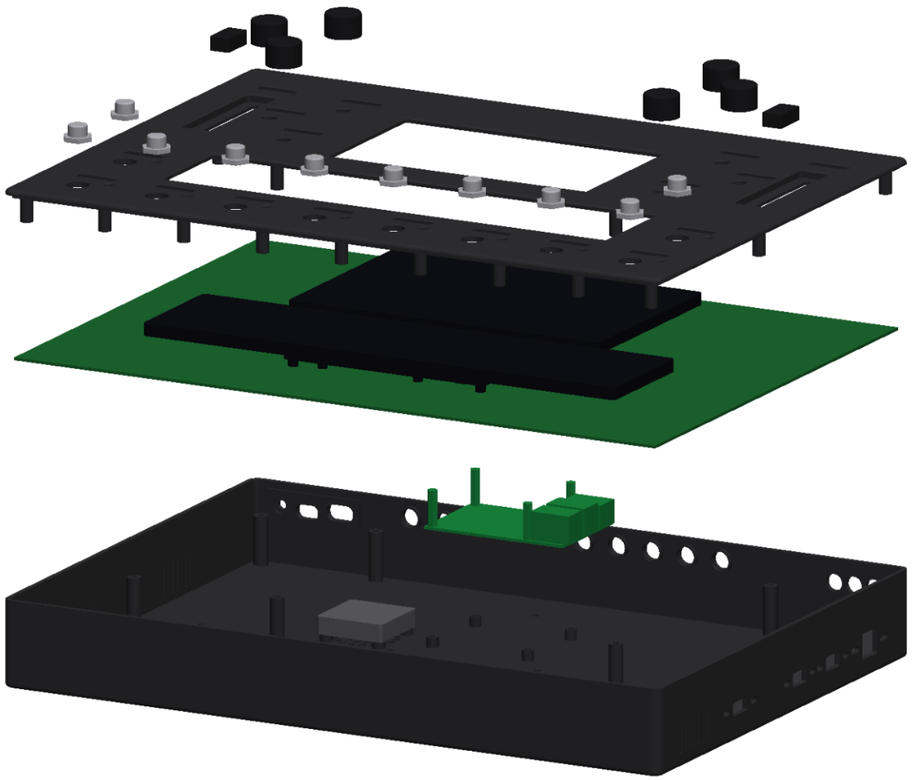
OVERALL SIZE 400 × 288 mm footprint, 49 mm base + 3 mm plate (52 mm) + feet; knobs stand 15 mm above the plate.

MATERIAL PETG / ASA / ABS, black. All tiles fit a 220 × 220 mm print bed.

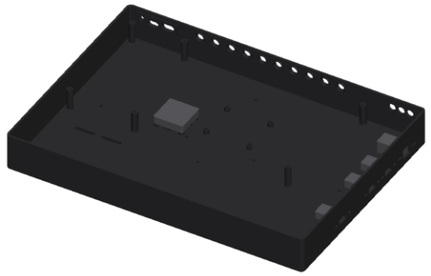
ORIGIN for every drawing: the outside FRONT-LEFT corner of the box, X to the right, Y toward the back.

ASSEMBLY SEQUENCE

Build it in this order. Steps 3-4: the screens screw on from the BACK of the board, so they go on before the board drops into the base.

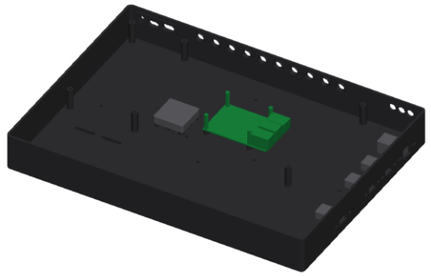


exploded: base, fan, Pi, board + screens, top plate, knobs



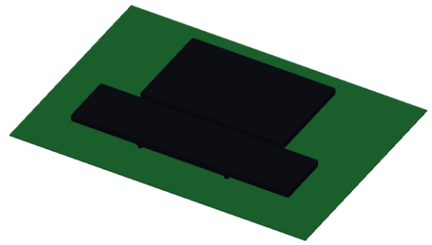
1 Base

Heat-set inserts (soldering iron, ~230 °C): 4 x M2.5 in the Pi bosses, 4 x M3 in the fan pads.
Fan onto the 4 floor pads, label down (blows out through the floor grille): 4 x M3 × 16.
4 panel-mount leads into the right wall, M3 × 8 + nuts. Power plunger into its back-panel hole from inside. 2S battery on its two straps.



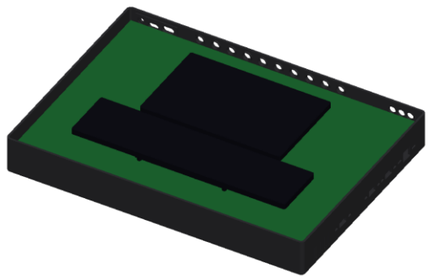
2 Raspberry Pi

Pi component side up, USB / Ethernet toward the right wall. 4 x M2.5 × 20 male-female standoffs: male end down through each Pi hole into the base inserts - this screws the Pi to the case.
Push the 2×20 stacking header onto the GPIO pins. Plug the side-panel leads into Ethernet and the two blue USB 3.0 ports, the HDMI lead into HDMI1 (by the audio jack). Plug in the bar-screen HDMI lead (HDMI0, by USB-C) and the USB-A to PH-4 lead (a black USB 2.0 port); leave their other ends loose.



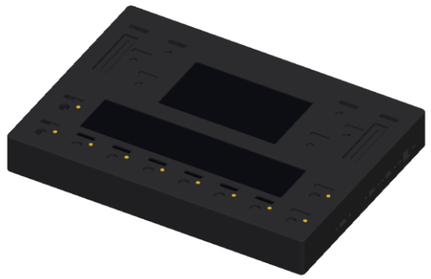
3 Screens onto the board

On the bench, BEFORE the board goes in - their screws go in from the back of the board.
7" display and 11.9" bar screen glass up, in their silkscreen outlines, 4 x M2.5 screws each from the back. Feed the 7" ribbon down through the slot beside J1. Display power lead → J_DISP, bar touch lead → J_USB_BAR (both on the back).



4 Board onto the Pi

Hold the board just above the base and connect: 7" ribbon → Pi DISPLAY port, HDMI lead → bar screen (through its slot), USB lead → J_USB_UP, fan → J_FAN, battery → J_BAT.
Lower it: J1 slides over the stacking header and the board settles on the 4 Pi standoffs and the 17 base bosses. Nothing should be pinched.



5 Top plate

Black foam tape around the two screen windows on the plate's underside (1 mm for the 7", 2 mm for the bar).
Set the plate on: shafts, levers, LEDs and footswitch bushings come through. Turn the pedal over and drive 17 x M3 × 40 from the bottom: base boss → board → plate insert. Snug, not tight.

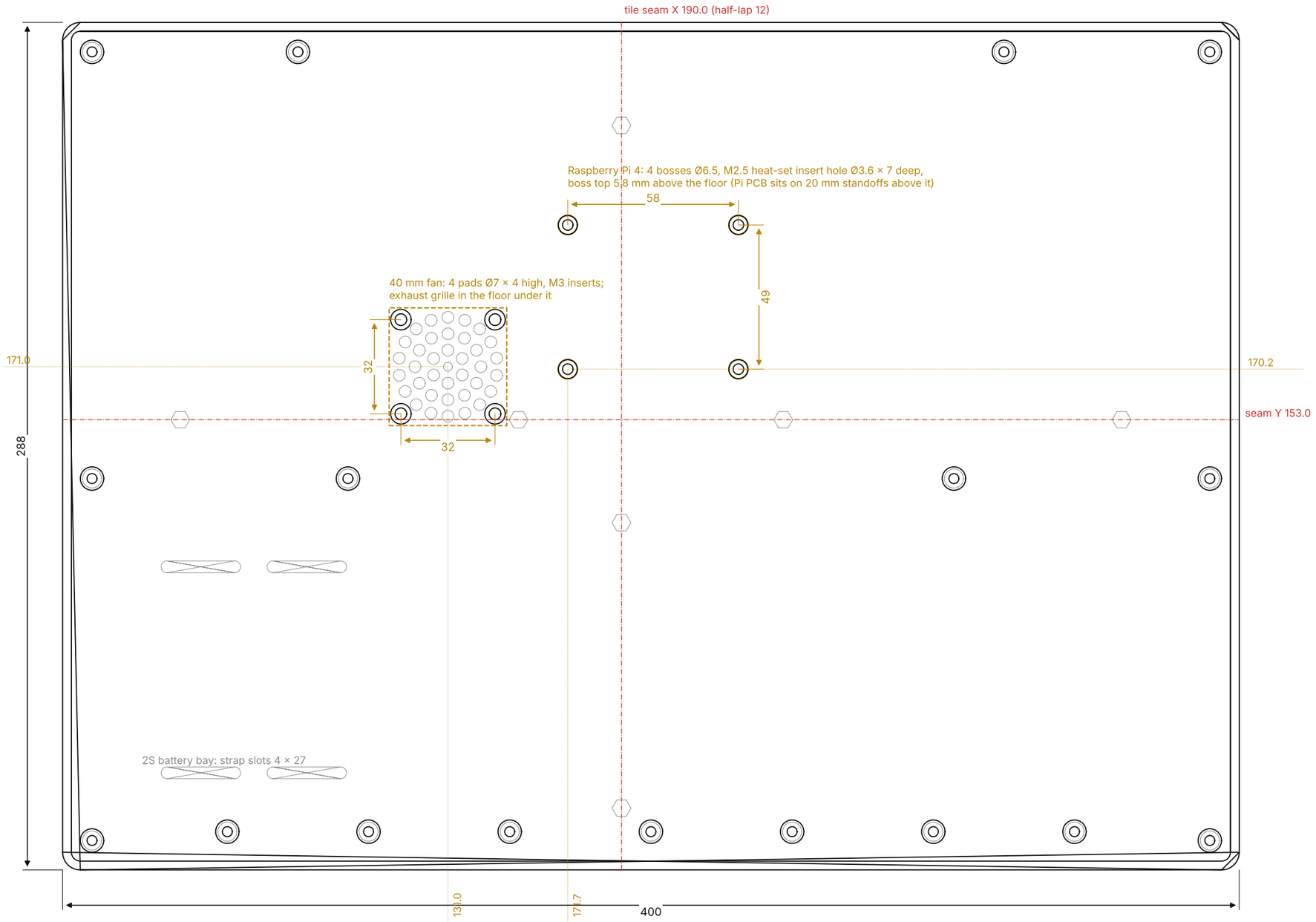


6 Finish

Washer + nut on each 1/4" jack outside the back panel (bagged with the board). Footswitch nuts on top. Push on the 6 knobs and 2 fader caps. 4 rubber feet in the corner recesses.
Power on with the button: the screens and label screens come up.

BASE - TOP VIEW

Section just above the floor (bold) and through the floor (thin): walls, bosses, Pi mounts, fan pads, grille, strap slots, seams.



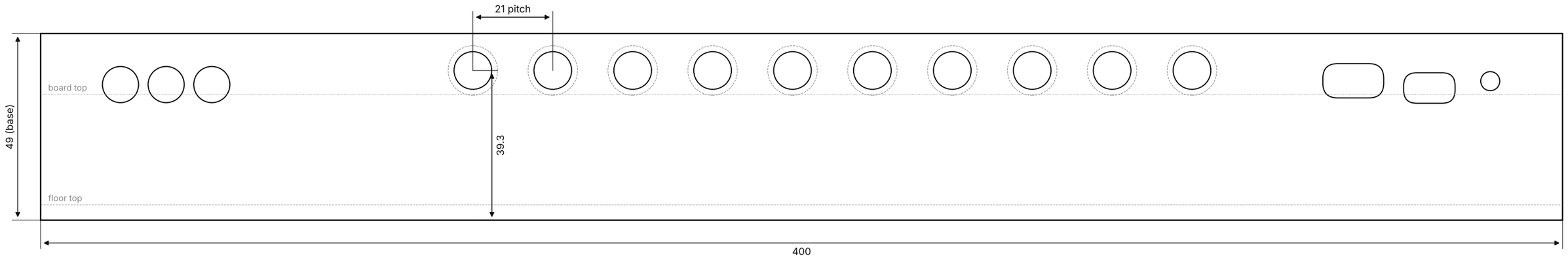
feature	X	Y	size
board boss 1	10.0	278.0	Ø8, M3 clear.
board boss 2	303.0	133.0	Ø8, M3 clear.
board boss 3	56.0	13.0	Ø8, M3 clear.
board boss 4	104.0	13.0	Ø8, M3 clear.
board boss 5	152.0	13.0	Ø8, M3 clear.
board boss 6	200.0	13.0	Ø8, M3 clear.
board boss 7	248.0	13.0	Ø8, M3 clear.
board boss 8	296.0	13.0	Ø8, M3 clear.
board boss 9	344.0	13.0	Ø8, M3 clear.
board boss 10	390.0	278.0	Ø8, M3 clear.
board boss 11	10.0	10.0	Ø8, M3 clear.
board boss 12	390.0	10.0	Ø8, M3 clear.
board boss 13	80.0	278.0	Ø8, M3 clear.
board boss 14	320.0	278.0	Ø8, M3 clear.
board boss 15	10.0	133.0	Ø8, M3 clear.
board boss 16	390.0	133.0	Ø8, M3 clear.
board boss 17	97.0	133.0	Ø8, M3 clear.
Pi boss 1	171.7	219.2	Ø6.5, M2.5 ins.
Pi boss 2	229.7	219.2	Ø6.5, M2.5 ins.
Pi boss 3	171.7	170.2	Ø6.5, M2.5 ins.
Pi boss 4	229.7	170.2	Ø6.5, M2.5 ins.
fan pad	115.0	155.0	Ø7, M3 insert
fan pad	115.0	187.0	Ø7, M3 insert
fan pad	147.0	155.0	Ø7, M3 insert
fan pad	147.0	187.0	Ø7, M3 insert
foot recess	20.0	268.0	Ø20 x 1
foot recess	380.0	268.0	Ø20 x 1
foot recess	20.0	20.0	Ø20 x 1
foot recess	380.0	20.0	Ø20 x 1

Floor 4 mm, walls 3 mm, corner R6,
bottom edge fillet R2.5. Bosses hold the board's
underside at Z 0 (floor top = -29).

BACK PANEL AND RIGHT SIDE PANEL

Seen from outside. Heights from the bottom of the base. All holes are cut in the base walls.

BACK PANEL (from behind)



item	X from right	height	hole
1/4" jack	302.6	39.3	Ø9.9 (+ ring 10.6/13 inside)
1/4" jack	113.6	39.3	Ø9.9 (+ ring 10.6/13 inside)
1/4" jack	281.6	39.3	Ø9.9 (+ ring 10.6/13 inside)
1/4" jack	134.6	39.3	Ø9.9 (+ ring 10.6/13 inside)
1/4" jack	260.6	39.3	Ø9.9 (+ ring 10.6/13 inside)
1/4" jack	239.6	39.3	Ø9.9 (+ ring 10.6/13 inside)
1/4" jack	218.6	39.3	Ø9.9 (+ ring 10.6/13 inside)
1/4" jack	197.6	39.3	Ø9.9 (+ ring 10.6/13 inside)
1/4" jack	176.6	39.3	Ø9.9 (+ ring 10.6/13 inside)
1/4" jack	155.6	39.3	Ø9.9 (+ ring 10.6/13 inside)
3.5 mm jack	45.0	35.6	Ø9.5
3.5 mm jack	33.0	35.6	Ø9.5
3.5 mm jack	21.0	35.6	Ø9.5
USB-C window	365.0	34.7	13.5 x 8.0, R3.4
USB-A window	345.0	36.6	16.0 x 9.0, R3.9
power plunger	381.0	36.5	Ø5.0

port	Y from front	height	cut-out / screw slots
RJ45	241.0	17.7	17.0 x 14.5; slots 27-38 c/c, 3.4 wide
USB	199.0	17.7	15.4 x 8.2; slots 26-32 c/c, 3.4 wide
USB	161.0	17.7	15.4 x 8.2; slots 26-32 c/c, 3.4 wide
HDMI	93.0	17.7	16.2 x 7.6; slots 28-34 c/c, 3.4 wide
vents	43.0	17.0	8 slots 2 x 18 at 5 pitch

RIGHT SIDE PANEL (from the right; front of the pedal at left)



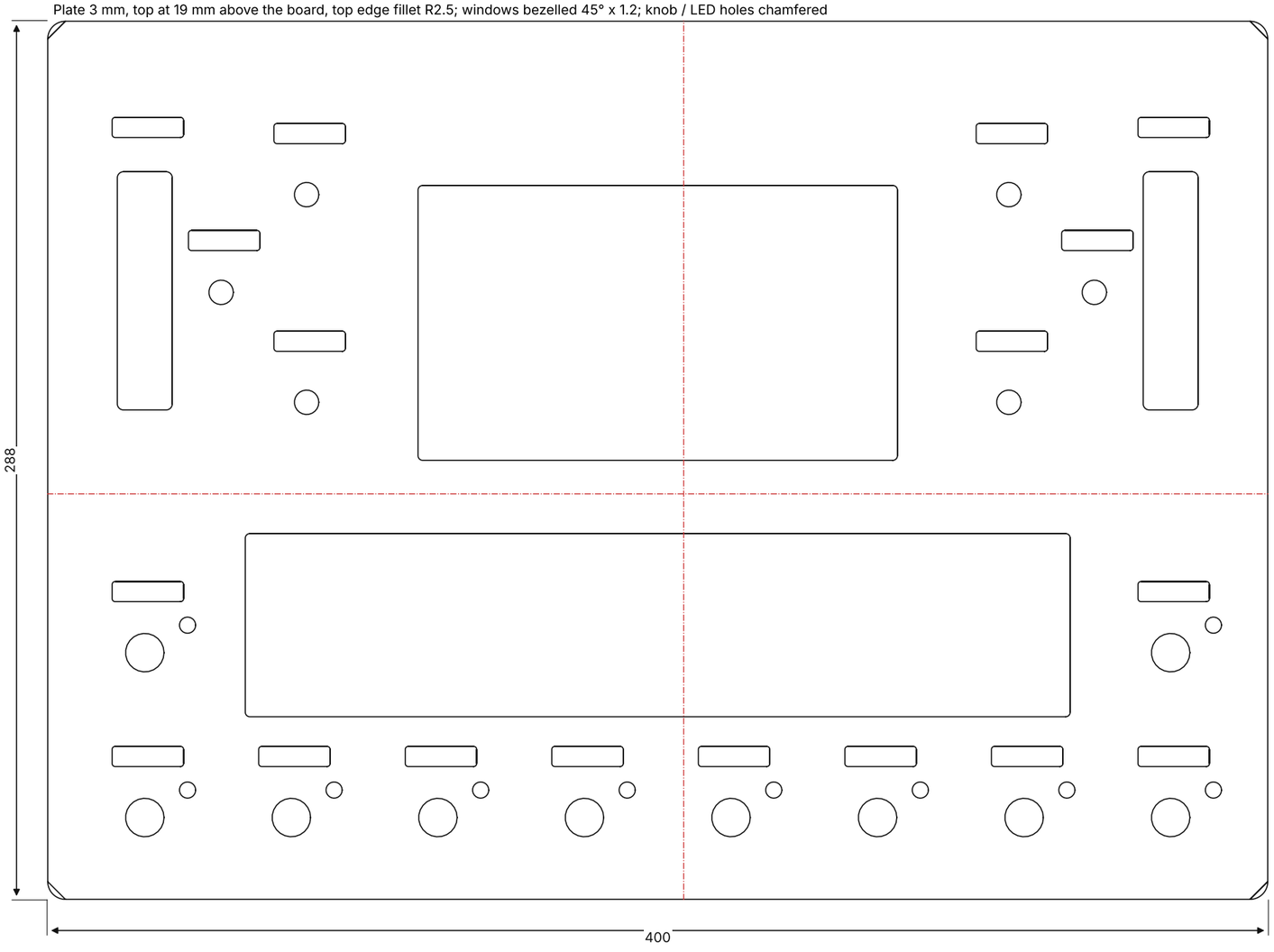
CENTRAL PEDAL rev B.3 MotionAudio.org

Back panel and right side panel

units mm | 2026-11-08 | sheet 4 / 8

TOP PLATE - TOP VIEW

Section through the plate (bold); dashed: tile seams. Coordinates of every opening on the right.

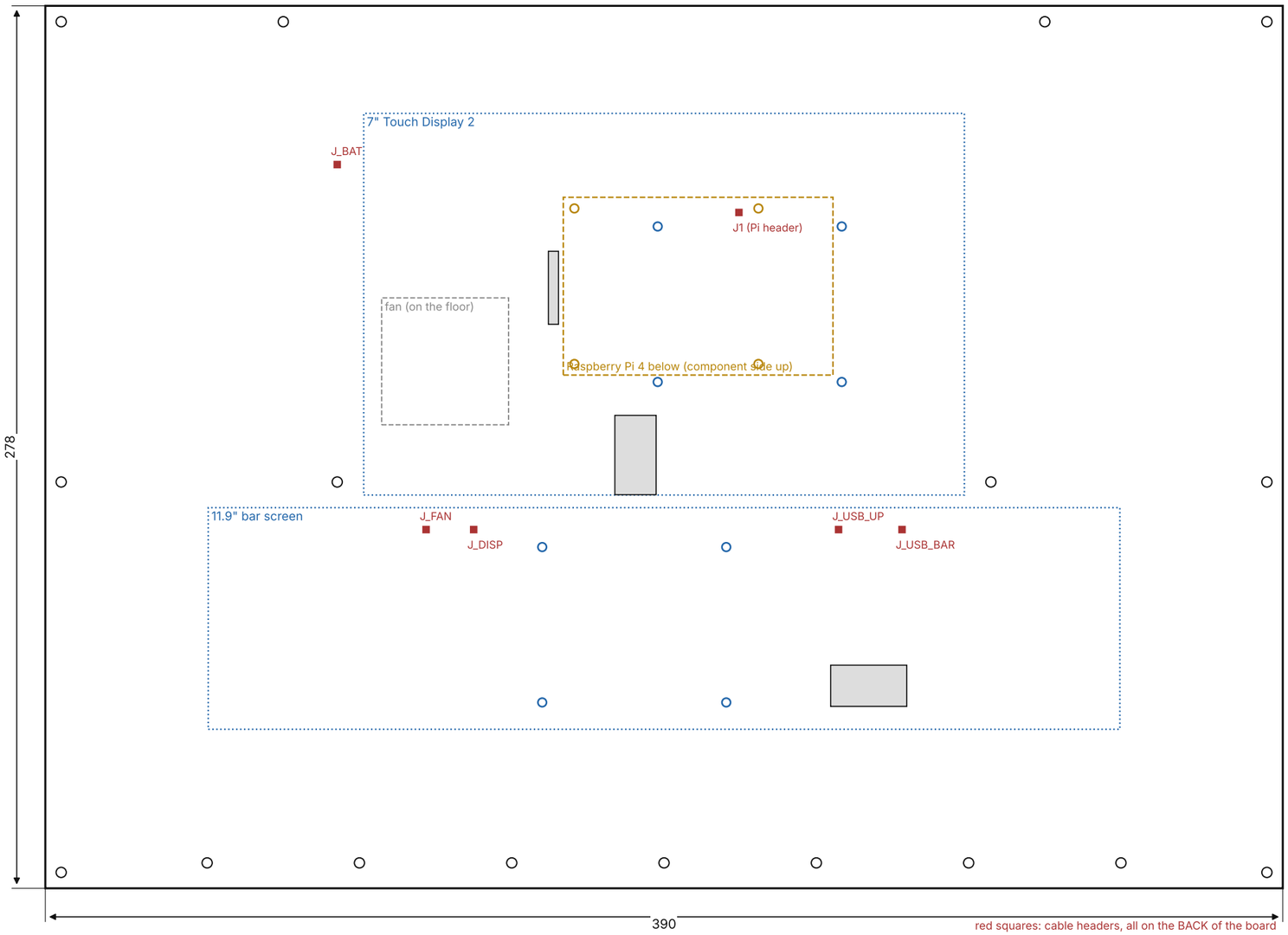


opening	X	Y	size	opening	X	Y	size	opening	X	Y	size
footswitch 1	32.0	27.0	Ø12.6	knob 3	85.0	163.0	Ø8.0	label K5	316.0	183.0	23.4 x 6.6
footswitch 2	368.0	81.0	Ø12.6	knob 4	315.0	231.0	Ø8.0	label K6	344.0	216.0	23.4 x 6.6
footswitch 3	80.0	27.0	Ø12.6	knob 5	315.0	163.0	Ø8.0	7in window	200.0	189.0	157.0 x 90.0
footswitch 4	128.0	27.0	Ø12.6	knob 6	343.0	199.0	Ø8.0	bar window	200.0	90.0	270.2 x 60.0
footswitch 5	176.0	27.0	Ø12.6	fader A	32.0	199.5	slot 3.6 x 64	boss 1	10.0	278.0	Ø7, M3 insert
footswitch 6	224.0	27.0	Ø12.6	fader B	368.0	199.5	slot 3.6 x 64	boss 2	303.0	133.0	Ø7, M3 insert
footswitch 7	272.0	27.0	Ø12.6	label A	33.0	253.0	23.4 x 6.6	boss 3	56.0	13.0	Ø7, M3 insert
footswitch 8	320.0	27.0	Ø12.6	label B	369.0	253.0	23.4 x 6.6	boss 4	104.0	13.0	Ø7, M3 insert
footswitch 9	368.0	27.0	Ø12.6	label FS1	33.0	47.0	23.4 x 6.6	boss 5	152.0	13.0	Ø7, M3 insert
footswitch 10	32.0	81.0	Ø12.6	label FS10	369.0	101.0	23.4 x 6.6	boss 6	200.0	13.0	Ø7, M3 insert
LED 1	46.0	36.0	Ø5.3	label FS2	81.0	47.0	23.4 x 6.6	boss 7	248.0	13.0	Ø7, M3 insert
LED 2	382.0	90.0	Ø5.3	label FS3	129.0	47.0	23.4 x 6.6	boss 8	296.0	13.0	Ø7, M3 insert
LED 3	94.0	36.0	Ø5.3	label FS4	177.0	47.0	23.4 x 6.6	boss 9	344.0	13.0	Ø7, M3 insert
LED 4	142.0	36.0	Ø5.3	label FS5	225.0	47.0	23.4 x 6.6	boss 10	390.0	278.0	Ø7, M3 insert
LED 5	190.0	36.0	Ø5.3	label FS6	273.0	47.0	23.4 x 6.6	boss 11	10.0	10.0	Ø7, M3 insert
LED 6	238.0	36.0	Ø5.3	label FS7	321.0	47.0	23.4 x 6.6	boss 12	390.0	10.0	Ø7, M3 insert
LED 7	286.0	36.0	Ø5.3	label FS8	369.0	47.0	23.4 x 6.6	boss 13	80.0	278.0	Ø7, M3 insert
LED 8	334.0	36.0	Ø5.3	label FS9	33.0	101.0	23.4 x 6.6	boss 14	320.0	278.0	Ø7, M3 insert
LED 9	382.0	36.0	Ø5.3	label K1	58.0	216.0	23.4 x 6.6	boss 15	10.0	133.0	Ø7, M3 insert
LED 10	46.0	90.0	Ø5.3	label K2	86.0	251.0	23.4 x 6.6	boss 16	390.0	133.0	Ø7, M3 insert
knob 1	57.0	199.0	Ø8.0	label K3	86.0	183.0	23.4 x 6.6	boss 17	97.0	133.0	Ø7, M3 insert
knob 2	85.0	231.0	Ø8.0	label K4	316.0	251.0	23.4 x 6.6				

Fader trays: 18 × 78 recess, floor 12.8 above the board, lever slot 3.6 × 64 through. Label windows 23.4 × 6.6 (X, Y = window centre). Screen windows: 45° bezel 1.2 mm. All X / Y from the outside front-left corner.

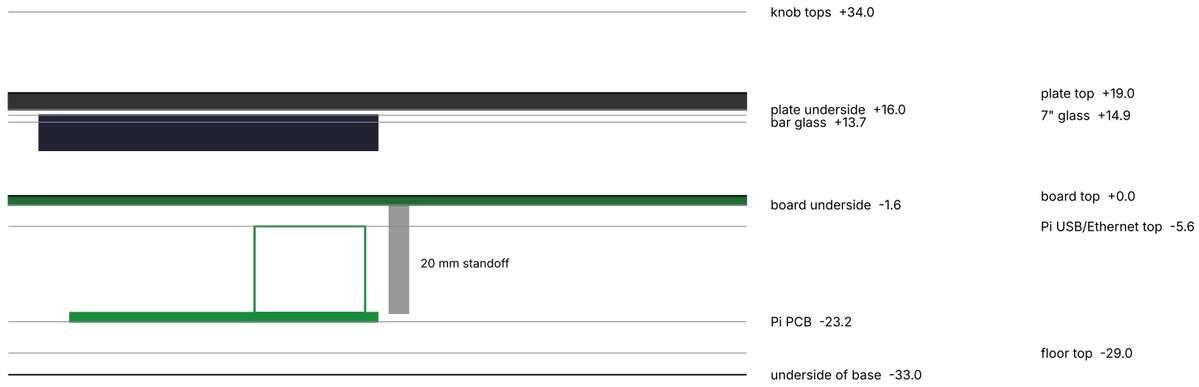
BOARD MECHANICAL AND HEIGHT STACK

Board coordinates (KiCad, origin top-left = back-left corner, Y toward the front). What screws where.



hole	x	y	use
Pi M2.5	166.74	63.82	rests on the 20 mm standoffs
Pi M2.5	224.74	63.82	rests on the 20 mm standoffs
Pi M2.5	166.74	112.82	rests on the 20 mm standoffs
Pi M2.5	224.74	112.82	rests on the 20 mm standoffs
7in M2.5	193.00	69.50	screw from the back
7in M2.5	251.00	69.50	screw from the back
7in M2.5	193.00	118.50	screw from the back
7in M2.5	251.00	118.50	screw from the back
bar M2.5	156.60	170.50	screw from the back
bar M2.5	156.60	219.45	screw from the back
bar M2.5	214.60	170.50	screw from the back
bar M2.5	214.60	219.45	screw from the back
slot	158.5-161.7	77.3-100.3	ribbon / plug cut-out
slot	247.5-271.5	207.7-220.7	ribbon / plug cut-out
slot	179.5-192.5	129.0-154.0	ribbon / plug cut-out
M3 x 17	see table	sheet 3	base boss -> board -> plate

HEIGHTS (mm, board top = 0)



Pi: 4 x M2.5 × 20 male-female standoffs, male end down through the Pi into the base inserts. The board's 4 Pi holes sit on the standoff tops; the 2×20 stacking header joins the Pi to J1, and the 17 M3 × 40 screws clamp base, board and plate together.

Screens: on their own M2.5 stand-offs, screwed from the BACK of the board - fit them on the bench before the board goes in. The 7" display covers the board's Pi holes, which is why those take no screws.

Gasket gaps plate-to-glass: 7" 1.08 mm, bar 2.35 mm.

WIRING AND CABLES

Every cable inside the box. The board's cable headers are all on its BACK; the Pi sits under it on the base floor.

RASPBERRY PI 4 (base floor)

DISPLAY (DSI) ← 7" ribbon

HDMI0, next to USB-C → bar screen

(= "HDMI 1" in the app: the app)

HDMI1, next to audio → right panel HDMI

(= "HDMI 2": projector lyrics)

USB 2.0 (black) → J_USB_UP

USB 3.0 (blue) x 2 → right panel

Ethernet → right panel

Power: 5 V from the board through the header

- never plug the Pi's own USB-C in

MAIN BOARD (headers on the back)

J1 2×20 socket ← stacking header

J_DISP PH 2 → 7" display power

J_USB_BAR PH 4 → bar screen touch

J_USB_UP PH 4 ← Pi USB (hub upstream)

J_FAN PH 2 → fan

J_BAT VH 2 ← 2S battery

Back panel (on the board): power button,

USB-C power/charge, USB-A, 10 × 1/4" jacks,

MIDI in / out, phones

SCREENS (screwed to the board)

7": ribbon → Pi DSI through the board slot;

power lead → J_DISP

11.9": HDMI ← Pi HDMI0; TOUCH micro-USB

← J_USB_BAR (touch + 5 V)

RIGHT SIDE PANEL / BASE

Ethernet, USB 3.0, USB 3.0, HDMI 2:

panel-mount leads from the Pi

Fan on the floor → J_FAN

2S pack on its straps → J_BAT

#	cable	from	to	wires / notes
1	2x20 stacking header	Pi GPIO pins	J1 on the board	push it onto the Pi first; the board's J1 slides over it
2	7" display ribbon (with the display)	display	Pi DISPLAY (DSI) port	through the board slot beside J1
3	7" display power lead	display	J_DISP	its own lead, plug swapped for a PH-2: red = pin 2 (5 V), black = pin 1
4	slim HDMI lead (mini-HDMI to micro-HDMI)	bar screen HDMI	Pi HDMI0 (next to USB-C)	plug body sits in the board's HDMI PLUG slot
5	bar screen USB lead (with the screen)	bar screen TOUCH	J_USB_BAR	USB-A end swapped for a PH-4: 1 red 5 V, 2 white D-, 3 green D+, 4 black GND
6	USB-A plug to PH-4 lead, ~20 cm	Pi USB 2.0 port	J_USB_UP	pin 1 empty, 2 white D-, 3 green D+, 4 black GND
7	RJ45 panel-mount lead	Pi Ethernet	right panel	M3 x 8 + nuts through the slotted ears
8	USB 3.0 panel-mount lead x 2	Pi USB 3.0 (blue)	right panel	M3 x 8 + nuts
9	micro-HDMI to HDMI panel-mount lead	Pi HDMI1 (next to audio)	right panel	projector / second screen
10	fan lead (PH-2 plug)	fan	J_FAN	red = pin 2 (5 V), black = pin 1
11	battery lead (VH-2 plug)	2S pack with BMS	J_BAT	pin 1 = battery -, pin 2 = battery + (check before plugging in)

PARTS TO BUY, HARDWARE AND PRINTING

Everything that is not on the PCBWay board. Cheap generic parts are fine where no brand is named.

qty	part	notes
1	Raspberry Pi 4 Model B (2 GB+)	powered from the board - never plug its own USB-C in
1	2x20 extra-long GPIO stacking header (~11 mm pins)	e.g. Adafruit 2223
4	M2.5 x 20 mm male-female hex standoff (6 mm thread)	Pi to base inserts; board rests on top
4	M2.5 heat-set insert (M2.5 x 4 x 3.5)	Pi bosses in the base
1	Raspberry Pi Touch Display 2 (7")	with its ribbon cables and screws
1	Waveshare 11.9inch HDMI LCD (touch, SKU 19454)	its screws; its USB cable becomes the touch lead
1	slim mini-HDMI to micro-HDMI lead, ~30 cm	bar screen to Pi HDMI0
1	short USB-A plug lead (~20 cm) + JST PH 4-pin plug	Pi USB 2.0 port -> J_USB_UP: pin 1 not used, 2 white D-, 3 green D+, 4 black GND
1	JST PH 2.0 2-pin + 4-pin pre-crimped pigtails, heat-shrink	display power and bar touch leads
1	40 x 40 x 10 mm 5 V fan, JST PH 2.0 plug	+ 4 x M3 x 16 screws, 4 x M3 heat-set inserts
1	RJ45 panel-mount extension lead, 20-30 cm	M3 ears, 27-38 mm apart
2	USB 3.0 A panel-mount extension lead, 30 cm	M3 ears, 26-32 mm apart
1	micro-HDMI male to HDMI-A female panel-mount lead, 30-50 cm	M3 ears, 28-34 mm apart
8	M3 x 8 screws + nuts	panel-mount leads
1	2S Li-ion pack with BMS, JST VH 2-pin lead	fits 80 x 70 mm bay, 2 x 25 mm hook-and-loop straps
21	M3 heat-set inserts (M3 x 5.7 x 4)	17 plate bosses + 4 splice-bar bosses
17	M3 x 40 cap screws	base -> board -> plate
4 / 7	M3 x 8 screws / M3 x 12 + nuts	splice bars / base tile laps
6	knobs for 6 mm D shafts, 18-20 mm	black with a gold line
2	fader caps for a 4 mm lever	
4	rubber feet Ø20	
1	black foam tape 1 mm and 2 mm, 5 mm wide	gaskets around the screen windows

PRINTING

PETG, ASA or ABS, black (matte if you have it). 0.2 mm layers. All parts fit a 220 × 220 mm bed.

Plate tiles: upside down (flat top on the bed), 5 perimeters, 40 % infill, 100 % top/bottom near the fader trays and lap steps.

Base tiles: floor down, 4 perimeters, 25 % infill. Back / side panel holes bridge without supports.

Splice bars, power plunger: flat, 100 % infill. Inserts: soldering iron at ~230 °C (445 °F), press straight, let cool.

Base tiles: glue the half-laps and bolt with the 7 x M3 × 12; plate tiles: glue the laps, screw the 2 splice bars.